

SymPy Bug Card

Bug 56: Singularities misses loggamma pole at zero

Task. Find the singularities of $\text{loggamma}(x)$ over the complex domain.

$\text{singularities}(\text{loggamma}(x), x, \mathbb{C})$

SymPy's answer.

$\text{singularities}(\text{loggamma}(x), x, \mathbb{C}) = \emptyset$

Correct answer. The singularity set must contain 0; more generally it contains the nonpositive integers $\{0, -1, -2, \dots\}$.

Explanation. $\text{loggamma}(0)$ evaluates to ∞ , and the recorded right-hand limit satisfies $\lim_{x \rightarrow 0^+} \text{loggamma}(x) = \infty$. Thus $x = 0$ is an actual pole inherited from $\Gamma(x)$, not a removable point or an equivalent empty-set representation.

Likely root cause. The diagnosis traces the call to `sympy/calculus/singularities.py`, where `singularities` scans only selected structural patterns such as `Pow` and `log` atoms; `loggamma(x)` is not rewritten to `log(gamma(x))` and has no special branch. This is a new failure mode with no public precedent.

Artifact (GitHub).

[bug-report/bug-056-singularities-misses-loggamma-pole-at-zero/](#)